Tong-Nong Lin

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Education

University of Texas at Austin – PhD student in Computer Engineering, GPA: 3.95/4.0
National Taiwan University – MS in Electrical Engineering
National Taiwan University – BS in Electrical Engineering and Mathematics

Experience

Research Assistant, UT Austin – Austin, TX Aug 2026 – Present

- LLM-assisted formal verification for `s2n-bignum`, AWS's HOL Light–verified cryptographic library.

Applied Scientist Intern, Amazon – Bellevue, WA May 2026 – Aug 2026

- Optimized an existing GPU implementation of a large-scale optimization solver to run **12x faster**, reaching **27x the CPU baseline**, and cutting a multi-hour solve down to minutes on a multi-GPU cluster.
- Cut time per iteration by **87%** after profiling showed most of it went to moving data rather than doing math: kept the **CSR** matrices and the warm-start vector resident on the GPU instead of re-uploading them on every solve, then halved memory traffic in the bandwidth-bound **sparse matrix-vector multiply** by switching it to **FP32**, changing the final objective by less than 0.05%.
- Followed the bottleneck as it shifted to the single-threaded coordinator: offloaded its vector math to a GPU with **CuPy**, verified the results stayed bit-identical, and stopped it from re-saving and re-loading its own state every iteration, cutting coordinator time **72%**.

Research Assistant, UT Austin – Austin, TX Aug 2024 – May 2026

- Achieved 2.12x performance improvement over Gunrock (established CUDA-based graph processing benchmark) by implementing optimized parallel algorithms in PyKokkos with selective output buffering, memory coalescing, and workload balancing techniques that eliminate redundant memory operations in GPU computing.
- Applied advanced parallel programming techniques including shared memory optimization, thread block configuration, and memory access pattern analysis to design high-performance GPU algorithms for graph traversal (BFS, SSSP) and centrality computation.
- Used profiling tools (NVIDIA Nsight Compute) to identify performance bottlenecks and optimize kernel execution efficiency in GPU-accelerated graph algorithms.

Research Assistant, Academia Sinica – Taipei, Taiwan Mar 2023 – Mar 2024

- Researched streaming algorithms for graph problems with sublinear space complexity, processing large-scale graphs using limited memory.
- Designed novel deterministic streaming algorithm to find independent sets meeting Turán's Bound.
- Developed deterministic algorithms for integer decomposition problems using probabilistic method for existence proofs, publishing results at IWOCA 2024 with provable time and space complexities.

Software Engineer, Mediatek – Hsinchu, Taiwan Sep 2019 – Mar 2023

- Developed and integrated **AES256-GCM cryptographic algorithms** into 5G modem software, ensuring secure data transmission compliant with 3GPP specifications.
- Implemented **authentication protocols and encryption systems** in boot mode without RAM usage for secure startup.
- Enhanced SIM-lock authentication protocols between mobile devices and carrier servers, implementing challenge-response mechanisms and certificate validation.
- Developed certificate framework supporting ASN.1 DER format parsing and validation, integrating with public key infrastructure for secure communications.
- Collaborated with cross-functional teams (hardware, QA, product management) in Agile development environment, delivering features on schedule for production chipsets.

Research Assistant, National Taiwan University – Taipei, Taiwan Mar 2017 – Mar 2019

- Conducted research in algorithmic game theory and mechanism design, focusing on risk aversion under uncertainty.

- Proposed generalized mathematical formulas modeling player risk aversion in multi-agent systems.
- Proved tight and near-tight upper and lower bounds on price of anarchy and Nash-equilibrium analysis.

Publications

[2] Aditya Thimmaiah, **Tong-Nong Lin**, Milos Gligoric, **MGQL: An Executable, Small-Step Semantics of GQL**, Proceedings of the ACM on Programming Languages (PACMPL), OOPSLA 2026

- Built the first mechanized, **small-step operational semantics** for a substantial read-only fragment of **ISO GQL** (ISO/IEC 39075), whose semantics the standard defines only informally across 600+ pages.
- Designed a **schema-aware type system** that refines variable types through closed-graph schemas and tracks nullability, and proved its **soundness**, guaranteeing well-formed queries return results conforming to declared schemas.
- Carried out the mechanization in **Lean 4**, yielding an **executable** semantics that runs queries directly from the formal rules and gives the first machine-checked bridge to the standard's informal specification.

[1] Tong-Nong Lin, Yu-Cheng Lin, Cheng-Chen Tsai, Meng-Tsung Tsai and Shih-Yu Tsai, **Efficient Algorithms for Decomposing Integers as Sums of Few Tetrahedral Numbers**, Proceedings of the 35th International Workshop on Combinatorial Algorithms (IWOCA), 2024

- Developed algorithms for **integer decomposition** improving theoretical bounds using **probabilistic and deterministic methods**.
- Analyzed time and space complexity with formal proofs and implemented efficient **C++** prototypes.

Projects

Program Analysis and Compiler Development

- Modified **GCC compiler** to support new language expressions.
- Used **ANTLR4** for lexing, parsing, and semantic analysis of **Trino SQL**.
- Leveraged **Java Pathfinder (JPF)** to build custom **memoization and code coverage tools** via listener mechanisms.
- Extended **OpenJDK** to support new syntax forms for expression grouping: [[Expression, Expression, Expression, ...]].

Services

OOPSLA '26 - sub-reviewer

Presentations

2026 AMD iMAGiNE Symposium.

Skills

Programming: C/C++(Proficient), Java, Python, Go, Rust, Lean, HOL Light
Frameworks: PyKokkos, CUDA, CuPy, Ray, OpenMP, ANTLR4, JPF, LLVM, Docker, Git, CMake, Linux/Unix
Expertise: Theorem Proving, Algorithm Design, Compiler, Functional Programming, Program Analysis, Parallel Programming, multithreaded programming, ADMM, Distributed Optimization, Performance Optimization, GPU Programming, Distributed System, Static Analysis, Graph Algorithms, Software Testing, High-Performance Computing (HPC), Nsight Compute, profiling tool
Databases: Neo4j, Graph Databases, SQL

Honors & Awards

Graduate School Fellowship, University of Texas at Austin 2024,2025,2026
Research Publication, IWOCA International Conference 2024